

The Feynman Grand Prize, Reframed

An XPRIZE-grade, material-agnostic specification for productive nanosystems

A proposed reframe of the Feynman Grand Prize for discussion — not an official Foresight/XPRIZE rule change.

The ask. The original Foresight Feynman Grand Prize (Freitas & Merkle, 1995) is a superb north-star but, judged as an XPRIZE-style competition, it is (1) underspecified — no defined measurement methods, judging protocol, operating environment, input classes, time-to-claim, or partial-credit structure — and (2) over-indexed on diamondoid: its implied chemistry (0.1 nm positional placement, bumps on an atomically-precise surface) presumes diamondoid / germanium-carbon mechanosynthesis, narrowing the solution space and over-weighting the single hardest chemistry. This reframe makes the prize objective, tiered, and material-agnostic — rewarding the capability, not the chemistry.

1. Design principles

- **Objective & measurable.** Every criterion has a defined metrology, a numeric pass/fail threshold, and independent verification. A judge can score it without trusting the entrant.
- **Material-agnostic.** Reward the verified positional construction of a SPECIFIED functional object — defined by geometry, function, and precision, not by composition. Any chemistry qualifies (single-site transition-metal catalysis, II–VI, oxides, carbon, biologically-templated inorganic, ...) if the product meets spec. Single-site catalysis is the generality mechanism; diamondoid is one allowed path, not the requirement.
- **Tiered (partial credit).** A ladder of objectively-scored tiers (precision → force/energy → generality → size) so intermediate productive-nanosystem capability wins real prizes. The original all-or-nothing bar becomes the top tier, not the entry gate — which both aligns incentives with the real technology path and makes the prize winnable in stages.
- **Fair & reproducible.** Keep the strong elements of the original: destructive testing of 32 copies (good statistics), self-replication NOT required, off-board power/control admissible. Add the missing specs: operating environment, input-signal classes, time-to-claim, and a named judging body with a published metrology protocol.

2. The reframed deliverables (output-defined)

Deliverable A — Positional construction engine. A nanoscale machine that builds a JUDGE-SPECIFIED target object to a stated positional tolerance, independently verified by metrology, while demonstrating a stated force capability and energy budget. The target is specified by geometry/function + precision; its composition is the entrant's choice. (This replaces “a 100 nm arm that positions to 0.1 nm” — a means — with “produce this verified object to this tolerance” — an end.)

Deliverable B — Molecular computing primitive. A nanoscale device that correctly computes 8-bit binary addition, demonstrated exhaustively over all 65,536 input pairs, with a readable output, within stated size and energy bounds (≤ 100 kT per bit-operation). Device physics and material are the entrant's choice; only correctness, size, energy, and readability are scored.

3. The tiered prize structure

Tier	Deliverable A — positional construction	Deliverable B — computing primitive	Proves (productive-nanosystem capability)
T1 — Entry	Build a specified object to ~ 1 nm tolerance, independently verified	8-bit adder correct (all pairs), ≤ 500 nm, readable output	Controllable, verified positional placement
T2 — Advanced	~ 0.3 nm tolerance; force capability ≥ 100 pN; stated energy budget	≤ 100 nm device	Sub-nm precision + mechanosynthesis-relevant force
T3 — Grand	~ 0.1 nm tolerance; atomically-precise specified product; force adequate to the chosen chemistry; 32 copies under destructive test	≤ 50 nm, ≤ 100 kT/bit-op, output on an atomically-precise surface	Atomically precise manufacturing — the original prize, now the top tier

Each tier is independently claimable and independently scored; an entrant need not attempt T3 to win T1/T2. Composition is unconstrained at every tier — a transition-metal-catalysed oxide product that meets the T2 tolerance scores identically to a diamondoid one.

4. Why material-agnostic matters

Diamondoid mechanosynthesis is one path, and the hardest one: stiff covalent C–C/Ge–C construction demands the largest positional forces and stiffness. Over-indexing on it both **narrows the field** (excluding entrants pursuing other valuable chemistries) and **over-weights the worst case**. A capability/output prize is fairer, more likely to be won, and more useful: it rewards any genuine productive nanosystem — a machine that positions matter precisely and verifiably — regardless of what it builds with. Single-site catalysis plus precision can, in principle, address much of the periodic table; the prize should reward reaching that capability, not one chemistry's version of it.

5. How this maps to NDI's program (context)

NDI's generational platform reaches the tiers in order: the closed-loop molecular arm (Gen-1, ~1–2 nm) is near T1; protein stiffening (Gen-2/3) targets T2's sub-nm and force; atom-by-atom construction (Gen-4+) is T3. The prize tiers and the development path are deliberately aligned: each prize tier corresponds to a generation of capability.

Bottom line for the prize. Keep the spirit (a self-contained nanoscale construction engine + a molecular computer, destructively verified across copies) but specify it for objective competition and free it from any single chemistry. The result is winnable in stages, fair across materials, and a faithful test of productive-nanosystem capability rather than of diamondoid in particular.